

Block to Python Bridge

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Requests Responses and JSON

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Conversation State and Memory

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Model Memory vs Retrieval

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Predict then observe for From a paper algorithm to Python. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for From a paper algorithm to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Set total=0, then for values 2,4,9 perform total=total+value. Trace the result. What changes if total=0 moves inside the loop before that addition?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

6. Solve the given case

A block program sets total=0 and adds each value in [2,4,6].

Translate and trace.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Trace this Python: `score = 2`; then if `score >= 2`: `score = score + 3`. What is the final value, and what changes if the condition becomes `score > 2`?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 23 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Implement small transparent functions for selection, validation, and citation formatting.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Connect the paper steps to Python

Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

My reasoning and evidence

13. W2 Predict then observe

Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual output, any error and what changed. If working on paper, label both results as traces.

My reasoning and evidence

14. W3 Explain an order error

Move `print(total)` to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

My reasoning and evidence

15. W4 Transfer the procedure independently

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using `trays`, `per_tray` and `tiles`. Predict its output. Then change only `trays` to 0 and predict again. Explain why adding the two quantities is wrong.

My reasoning and evidence

16. W5 Keep an honest conclusion

Explain the difference between a successful run and a correct calculation. State one limit of this program and what you would keep so another person can check your work.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Block to Python Bridge

Translate a “count qualifying books” paper procedure into a Python-style trace.

Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Block to Python Bridge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Start counter

count begins at 0 before the loop.



2. Trace decisions

80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2.



3. Report count

The final result is 2 qualifying books, not the sum of their page counts.

CHANGE ONE THING

Replace the last page count 90 with 100.

Explain the changed result using the matching lesson.

Trace the changed list and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Translate a “count qualifying books” paper procedure into a Python-style trace.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final result is 2 qualifying books, not the sum of their page counts.

1. Trace decisions
2. Start counter
3. Report count

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Report count The final result is 2 qualifying books, not the sum of their page counts. Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Replace the last page count 90 with 100. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The final count becomes 1.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The final count becomes 1. The comparison is strictly less than 100, so the boundary value does not qualify.

Repair a missing building block

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records. Short practice: Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Conversation State and Memory (ai-3/20). Goal: Explain the mechanism and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the mechanism and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
